

Multi-Role Buffer Protocol

Batch Manufacturing Record

Run: Run-2026-003 (With Figures) Status: ACTIVE Started: Completed:

Project: Protocol Version: v2 Generated: April 1, 2026

1. General Information

Product Name / Candidate	Multi-Role Buffer Protocol
Batch / Lot Number	Run-2026-003 (With Figures)
Outcome	None
Process SOP / Protocol Reference	Multi-Role Buffer Protocol (v2)

2. Bill of Materials (BOM)

Record all raw materials, media, and buffers utilized in this run.

Material Description	Part / Lot Number	Required Quantity	Actual Quantity Added	Operator Initials & Date

3. Equipment Log

Record primary equipment used to ensure traceability.

Equipment Name / Type	Equipment ID (Asset Tag)	Calibration Status (Valid Until)	Operator Initials & Date

4. Execution: Unit Operations

Execute process steps. Record actual values, operator, and verifier (if required by SOP) for each step.

4.1 Media Prep

Step	Instruction	Target Parameter	Actual Value	Operator	Verifier
1	Weigh Reagents	Weigh NaCl and KCl. NaCl: 8, and KCl: 0.2.	Recorded: 8.01 2026-04-01T10:30:00Z	D.S.C.	
2	Dissolve	Add to 1000 mL water and stir.	Recorded: 1000 2026-04-01T10:45:00Z	D.S.C.	

4.2 QC

Step	Instruction	Target Parameter	Actual Value	Operator	Verifier
1	Measure pH	Measure and adjust pH. Target pH: 7.4.	Recorded: 7.38 2026-04-01T11:15:00Z	J.W.	

5. Deviations and Process Comments

Log any deviations from target parameters or SOP. Run notes and anomalies are surfaced here.

Step / Status Ref.	Description of Deviation / Observation	Impact Assessment Required? (Y/N)	Lead Reviewer Sign-off
ACTIVE	Observed slight turbidity after buffer prep.		Dr. Sarah Chen 2026-04-01 10:35:00
ACTIVE	[ANOMALY] ANOMALY: Cell viability below threshold at 85%.		Dr. Sarah Chen 2026-04-01 11:20:00

6. Final Disposition & Signatures

By signing below, the responsible parties certify that the material described herein was produced according to the specified process and that all deviations have been reviewed.

Protocol Approvals

Run Sign-offs

Wet-Ink Sign-Off

Role	Name (Print)	Signature	Date
Lead Operator / Engineer			
Process Development Lead			
Quality / Compliance (if applicable)			

Appendix A: Figures

Figure 1: test_figure_1.png (Weigh Reagents, uploaded 2026-04-01 10:32)

Figure 2: test_figure_2.png (Dissolve, uploaded 2026-04-01 10:46)

Figure 3: test_figure_3.png (Measure pH, uploaded 2026-04-01 11:16)